

Task 1 : LOCAL DATA ANALYSIS OF SYSTEM LOGS

1. GitHub Link

<https://github.com/16-alex-16/Internship-TraineeLand>

2. Tools and Libraries

The project was developed using Python with the following libraries:

- Pandas: used for structured data handling and transformation
- Matplotlib: used for data visualization (bar charts and pie charts)
- Evtx: used to read Windows Event Log (.evtx) files
- xml.etree.ElementTree: used to parse XML event data

3. Data Source

The input data consists of Windows Event Logs exported in .evtx format:

- Security logs (authentication and privilege events)
- User logon logs

Each event is parsed to extract:

Event ID	Timestamp	Username	IP address	Workstation name	Logon type	Status
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4. Data Processing

The logs are iterated using the Evtx library. Each record is converted from XML format into structured Python dictionaries. The data is then stored in a pandas DataFrame for further processing.

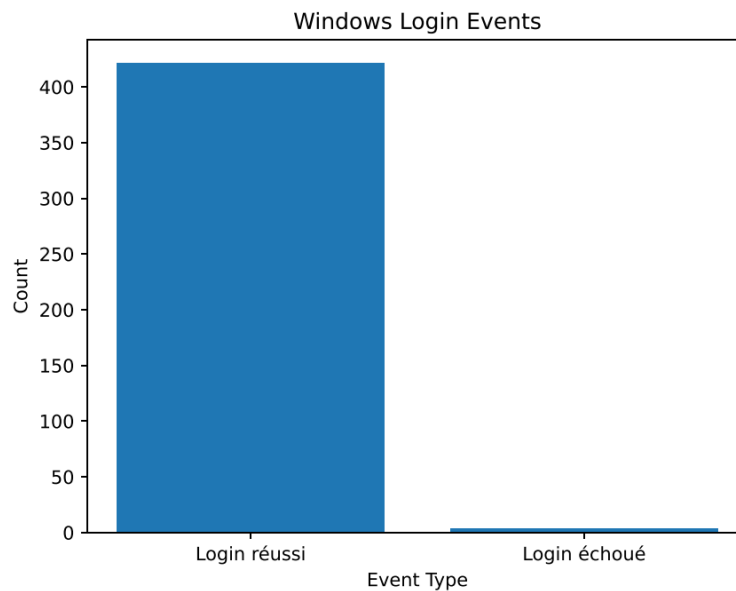
Event IDs are mapped to human-readable labels such as:

- 4624 → Login successful
- 4625 → Login failed
- 4672 → Admin privilege logon
- 4648 → Explicit credentials logon

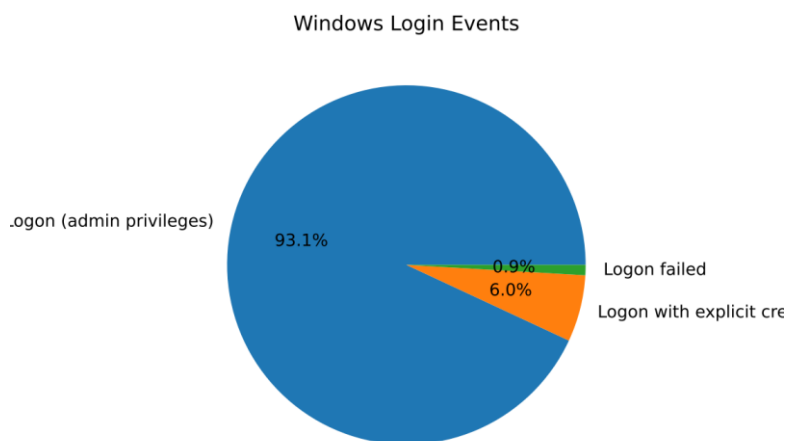
5. Data Visualization

Two types of visualizations were generated using matplotlib:

- Bar chart: distribution of user login events (successful, failed, etc.)



- Pie chart: distribution of security-related authentication events



6. Automation

The script is automated using Windows Task Scheduler. It is configured to run weekly without manual execution. This ensures continuous monitoring and periodic generation of updated analysis reports. A PDF file titled “Proof of automation with Task Scheduler” has been included to demonstrate the setup.

7. Output

The script generates PDF files containing visualizations:

- graph_users.pdf for user's logon
- graph_security.pdf for security logs